

Matrices In Geometry



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ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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iIn this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE-BM-2025

A. Formulae

2. ALGEBRA, LOGARITHMS AND COUNTING

A. Power-log swap identity

For any positive reals a, b :

$$a^{\log_{10} b} = b^{\log_{10} a}. \quad (0.1)$$

Derivation. Take $\log_{10}$ of the left side and use $\log(u^v) = v \log u$:

$$\log_{10}(a^{\log_{10} b}) = \log_{10} b \cdot \log_{10} a = \log_{10}(b^{\log_{10} a}). \quad (0.2)$$

Since $\log_{10}$ is strictly increasing (one-to-one), the arguments are equal. $\square$ Q.4

B. Digit constraints

A decimal digit lies in $\{0, \dots, 9\}$, so each digit of a number with digit-product 70 = $2 \cdot 5 \cdot 7$ is a divisor of 70 that is at most 9: one of 1, 2, 5, 7. A digit 1 would force the other two digits to multiply to 70, but the largest product of two allowed digits is $7 \cdot 7 = 49 < 70$. So no digit is 1; every digit is in $\{2, 5, 7\}$ and the product 70 (which contains each prime exactly once) forces the digits to be 2, 5, 7 in some order (by unique prime factorisation). Q.2

C. Sum of squares of consecutive integers

$$k^2 + (k+1)^2 = 2k^2 + 2k + 1. \quad (0.3)$$

This is increasing for $k \geq 0$; it is ≤ 999 iff $2k^2 + 2k \leq 998$, i.e. $k^2 + k \leq 499$, i.e. $k \leq 21$ ($21 \cdot 22 = 462 \leq 499 < 22 \cdot 23 = 506$). Q.9

D. Mean, median, mode

For an ordered sample $x_{(1)} \leq \dots \leq x_{(n)}$ with n odd:

$$\text{mean} = \frac{1}{n} \sum x_i, \quad \text{median} = x_{((n+1)/2)}, \quad (0.4)$$

and the mode is the value(s) of maximum frequency; a “single mode” means the maximum frequency is attained by exactly one value. Q.7

E. Binomial theorem and number of subsets

$$(1+x)^N = \sum_{k=0}^N \binom{N}{k} x^k. \quad (0.5)$$

Putting $x = 1$: $\sum_k \binom{N}{k} = 2^N$, the number of subsets of an N -set (including $\emptyset$); the number of non-empty subsets is $2^N - 1$.

Even-product counting. A product is odd iff all factors are odd, so for $S = \{1, \dots, 2026\}$ (with 1013 odd elements):

$$\#\{\text{non-empty subsets, even product}\} = (2^{2026} - 1) - (2^{1013} - 1) = 2^{1013}(2^{1013} - 1). \quad (0.6)$$

Q.19

F. Stars and bars via generating functions

The number of solutions of $n_1 + \dots + n_r = m$ in non-negative integers is the coefficient $[x^m](\sum_{n \geq 0} x^n)^r = [x^m](1-x)^{-r}$. From the binomial series

$$(1-x)^{-r} = \sum_{m \geq 0} \binom{m+r-1}{r-1} x^m, \quad (0.7)$$

which follows by differentiating $\frac{1}{1-x} = \sum x^n$ repeatedly ($\frac{1}{(r-1)!} \frac{d^{r-1}}{dx^{r-1}}$) or by counting arrangements of m stars and $r-1$ bars. Hence

$$\#\{n_i \geq 0\} = \binom{m+r-1}{r-1}, \quad \#\{n_i \geq 1\} = [x^m]x^r(1-x)^{-r} = \binom{m-1}{r-1}. \quad (0.8)$$

For $r = 4, m = 20$: $\binom{23}{3}$ and $\binom{19}{3}$. Q.20

G. Multiple binomial selections (independent entries)

If n independent Bernoulli(p) trials are made, the number of successes K satisfies

$$P(K = k) = \binom{n}{k} p^k (1-p)^{n-k}. \quad (0.9)$$

For $p = \frac{1}{2}$, $P(K = k) = \binom{n}{k}/2^n$. For entries in disjoint positions the events are independent, so probabilities multiply; for a shared entry we condition on its value and use the total probability theorem (??) below. Q.64

3. GEOMETRY AND CALCULUS

A. Inscribed angle theorem

An angle subtended at the circle by a chord is half the central angle subtended by the same arc:

$$\angle PQR = \frac{1}{2} \angle POR. \quad (0.10)$$

Derivation (case O on a side of the angle). Let QO meet the circle again at T . Triangle OPQ is isosceles ($OP = OQ = r$), so $\angle OQP = \angle OPQ = \alpha$ and the exterior angle at O is $\angle POT = 2\alpha$. Likewise $\angle ROT = 2\beta$ with $\beta = \angle OQR$. Then $\angle POR = 2(\alpha + \beta) = 2\angle PQR$. The other configurations follow by adding or subtracting. $\square$

B. Chord length, triangle and kite area

For a chord subtending central angle ϕ in a circle of radius r :

$$\text{chord} = 2r \sin \frac{\phi}{2}, \quad \phi = 90^\circ \Rightarrow PR = r\sqrt{2}. \quad (0.11)$$

Distance from the centre to the chord: $r \cos \frac{\phi}{2} = \frac{r}{\sqrt{2}}$. Area of a triangle = $\frac{1}{2}(\text{base})(\text{height})$.

Arrowhead area. If O lies inside triangle PQR and $QO \perp PR$, with h_Q, h_O the distances of Q, O from PR and $h_Q - h_O = QO$:

$$[PQRO] = [PQR] - [POR] = \frac{1}{2}PR(h_Q - h_O) = \frac{1}{2}PR \cdot QO. \quad (0.12)$$

For $r = 10$: $\frac{1}{2}(10\sqrt{2})(10) = 50\sqrt{2}$. Q.10

C. Subspaces, planes and discs

A 2-dimensional subspace of $\mathbb{R}^3$ is $\{x : a^T x = 0\}$ for a non-zero normal a , which always contains the origin. The ball $\{x^T x \leq R^2\}$ centred at the origin intersects such a plane in $\{y \in \text{plane} : \|y\| \leq R\}$, a disc of radius R and area πR^2 (here $R = 4$). Q.22

D. Factor theorem and roots of a cubic

If $f(a) = 0$ then $(x - a)$ divides f . For $f(x) = x^3 - 3x^2 + 2$: $f(1) = 0$ and polynomial division gives

$$f(x) = (x - 1)(x^2 - 2x - 2), \quad x^2 - 2x - 2 = 0 \Rightarrow x = \frac{2 \pm \sqrt{4 + 8}}{2} = 1 \pm \sqrt{3}. \quad (0.13)$$

E. Extrema of a differentiable function

At an interior extremum $f'(x_0) = 0$ (Fermat); the second derivative test says $f''(x_0) > 0$ gives a local minimum and $f''(x_0) < 0$ a local maximum. For $f = x^3 - 3x^2 + 2$:

$$f' = 3x^2 - 6x = 3x(x - 2), \quad f'' = 6x - 6, \quad f''(0) = -6 < 0, \quad f''(2) = 6 > 0. \quad (0.14)$$

On a half-open interval $(-1, 3]$ a global extremum can also occur at the included endpoint but never at an excluded one: $f(-1^+) \rightarrow -2$ is approached but not attained, so $\min f = f(2) = -2$ attained only at 2, while $f(0) = f(3) = 2$ gives two maximisers. Q.27

F. Geometric series

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad |a| < 1; \quad \sum_{n=1}^{\infty} a^n = \frac{a}{1-a}. \quad (0.15)$$

Derivation. $S_N = \sum_{n=0}^N a^n$ satisfies $S_N - aS_N = 1 - a^{N+1}$, so $S_N = \frac{1 - a^{N+1}}{1 - a} \rightarrow \frac{1}{1 - a}$. Q.35

G. Z-transform of a geometric sequence

$$\{a^n u[n]\}(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - az^{-1}}, \quad |z| > |a|. \quad (0.16)$$

Evaluating at $z = 1$ (inside the region of convergence when $|a| < 1$) gives the sum of the series; for the shifted sequence $a^n u[n-1]$ one gets $\frac{az^{-1}}{1-az^{-1}} \Big|_{z=1} = \frac{a}{1-a}$. This reproduces (??). For matrices, $\{M^k\} = z(zI - M)^{-1}$ (same derivation with $a \rightarrow M$), whose poles are the eigenvalues of M . Q.35, Q.21

4. LINEAR ALGEBRA

A. Eigenvalues, trace and characteristic polynomial

Eigenvalues are roots of $\det(A - \lambda I) = 0$. The coefficient of λ^{n-1} shows

$$\sum_i \lambda_i = \text{tr}(A), \quad \prod_i \lambda_i = \det A. \quad (0.17)$$

For a block-diagonal matrix the characteristic polynomial is the product of the blocks' polynomials. Q.46

B. Rotation matrix: eigenvalues and powers

For $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$:

$$\det(R - \lambda I) = (\cos \theta - \lambda)^2 + \sin^2 \theta = \lambda^2 - 2\lambda \cos \theta + 1 = 0 \Rightarrow \lambda = \cos \theta \pm i \sin \theta = e^{\pm i\theta}. \quad (0.18)$$

Powers. By the angle-addition formulas, $R(\alpha)R(\beta) = R(\alpha + \beta)$: the $(1, 1)$ entry is $\cos \alpha \cos \beta - \sin \alpha \sin \beta = \cos(\alpha + \beta)$, the $(2, 1)$ entry is $\sin \alpha \cos \beta + \cos \alpha \sin \beta = \sin(\alpha + \beta)$, etc. By induction

$$R(\theta)^n = R(n\theta). \quad (0.19)$$

Equivalently, with $R = P \text{diag}(e^{i\theta}, e^{-i\theta}) P^{-1}$, $R^n = P \text{diag}(e^{in\theta}, e^{-in\theta}) P^{-1}$. Since $R(\theta + 2\pi k) = R(\theta)$, only $n\theta \bmod 2\pi$ matters: for $\theta = \frac{2\pi}{5}$, $R^n = R^{n \bmod 5}$ and $2026 = 5 \cdot 405 + 1$. Q.21, Q.46

C. Centring matrix $M = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^T$

Using $\mathbf{1}^T \mathbf{1} = n$:

$$M^T = M, \quad (0.20)$$

$$M^2 = I - \frac{2}{n} \mathbf{1}\mathbf{1}^T + \frac{1}{n^2} \mathbf{1}(\mathbf{1}^T \mathbf{1})\mathbf{1}^T = I - \frac{1}{n} \mathbf{1}\mathbf{1}^T = M, \quad (0.21)$$

$$\text{tr}(M) = n - \frac{1}{n} \text{tr}(\mathbf{1}\mathbf{1}^T) = n - \frac{1}{n} \cdot n = n - 1, \quad (0.22)$$

$$M\mathbf{1} = \mathbf{1} - \frac{1}{n} \mathbf{1} n = \mathbf{0}, \quad Mv = v \text{ if } \mathbf{1}^T v = 0. \quad (0.23)$$

So M is a symmetric idempotent matrix, i.e. an orthogonal projection, with eigenvalue 0 (eigenvector $\mathbf{1}$) once and eigenvalue 1 with multiplicity $n - 1$ (the subspace $\mathbf{1}^\perp$); its rank equals its trace $= n - 1$. Q.52, Q.53, Q.65

D. Spectral theorem and Rayleigh quotient

A real symmetric A has orthonormal eigenvectors u_i with real eigenvalues λ_i . For $\|x\| = 1$ write $x = \sum c_i u_i$, $\sum c_i^2 = 1$:

$$x^T A x = \sum_i \lambda_i c_i^2 \leq \lambda_{\max} \sum_i c_i^2 = \lambda_{\max}, \quad (0.24)$$

with equality for $x = u_{\max}$. For M in (??), $\lambda_{\max} = 1$. Q.65

5. PROBABILITY AND STATISTICS

A. Total probability and Bayes' theorem

For a partition $\{D, \bar{D}\}$:

$$P(+) = P(+|D)P(D) + P(+|\bar{D})P(\bar{D}), \quad (0.25)$$

$$P(D|+) = \frac{P(+ \cap D)}{P(+)} = \frac{P(+|D)P(D)}{P(+)}. \quad (0.26)$$

The second line follows from $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$. Q.57, Q.64

B. Variance identities

$$(W) = [W^2] - (W)^2, \quad [UV] = U V \text{ for independent } U, V. \quad (0.27)$$

For independent X, Y with $Y = 0$: $[(g(X)Y)] = g(X) Y = 0$, so

$$(g(X)Y) = [g(X)^2] [Y^2]. \quad (0.28)$$

For $X \in \{0, 1\}$, $g(X) = 2X - 1 \in \{\pm 1\}$, thus $g(X)^2 \equiv 1$; also $[Y^2] = Y + (Y)^2$. Q.44

C. Pairwise-difference identity

With $\bar{x} = \frac{1}{n} \sum x_i$ and $d_i = x_i - \bar{x}$ (so $\sum d_i = 0$):

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2 &= \sum_{i,j} (d_i - d_j)^2 = \sum_{i,j} (d_i^2 + d_j^2 - 2d_i d_j) \\ &= 2n \sum_i d_i^2 - 2 \left(\sum_i d_i \right)^2 = 2n \sum_i (x_i - \bar{x})^2. \end{aligned} \quad (0.29)$$

Q.62

D. Total expectation, covariance, correlation

$$[Y] = [Y|X], \quad [XY] = [X[Y|X]], \quad (0.30)$$

$$(X, Y) = [XY] - X Y, \quad \rho = \frac{(X, Y)}{\sigma_X \sigma_Y}. \quad (0.31)$$

For $U \sim \text{Uniform}(a, b)$: $U = \frac{a+b}{2}$. For $X \sim \text{Uniform}(-1, 1)$: $[X^3] = \int_{-1}^1 \frac{1}{2} x^3 dx = 0$ (odd integrand, symmetric limits). Q.63

E. Exponential distribution and memorylessness

For mean λ : pdf $f(x) = \frac{1}{\lambda} e^{-x/\lambda}$, $x \geq 0$, survival $S(t) = \int_t^\infty f = e^{-t/\lambda}$. Then

$$P(X > s+t | X > s) = \frac{S(s+t)}{S(s)} = e^{-t/\lambda} = S(t). \quad (0.32)$$

Q.34

F. Gaussian integral and moment generating functions

Moment generating function $M_X(s) = [e^{sX}]$ (a two-sided Laplace transform of the pdf at $-s$); for independent X_i , $M_{\sum X_i} = \prod M_{X_i}$ and the MGF determines the law. χ_1^2 . For $X \sim N(0, 1)$, $s < \frac{1}{2}$:

$$[e^{sX^2}] = \int_{-\infty}^{\infty} \frac{e^{-(1-2s)x^2/2}}{\sqrt{2\pi}} dx = (1-2s)^{-1/2}, \quad (0.33)$$

using $\int e^{-ax^2/2} dx = \sqrt{2\pi/a}$ with $a = 1-2s$. Hence $\sum_{i=1}^n X_i^2$ has MGF $(1-2s)^{-n/2}$, which defines χ_n^2 .

Exponential. For $\text{Exp}(\text{mean } \theta)$:

$$[e^{sX}] = \int_0^\infty \frac{1}{\theta} e^{-x(1/\theta-s)} dx = \frac{1}{1-\theta s}. \quad (0.34)$$

With $\theta = 2$ this is $(1-2s)^{-1}$, so $\chi_2^2 = \text{Exp}(\text{mean } 2)$.

Sample mean. If X_i are i.i.d. $N(0, 1)$ then $\bar{X} \sim N(0, \frac{1}{n})$ (sum of independent Gaussians), so $\sqrt{n}\bar{X} \sim N(0, 1)$ and $(\sqrt{n}\bar{X})^2 \sim \chi_1^2$. Q.53

G. Cochran-type result for projections

Let $X \sim N(0, I_n)$ and M a symmetric idempotent matrix of rank r . Write $M = U\Lambda U^T$ with U orthogonal and Λ having r ones and $n-r$ zeros. Then $Y = U^T X \sim N(0, U^T U) = N(0, I_n)$ and

$$X^T M X = Y^T \Lambda Y = \sum_{k=1}^r Y_k^2 \sim \chi_r^2. \quad (0.35)$$

Applying this to the centring matrix ($r = n-1$) and noting $\sum (X_i - \bar{X})^2 = X^T M X$ gives χ_{n-1}^2 . Q.53

H. Properties of a discrete CDF

For a pmf $p(x_k) \geq 0$, $\sum p(x_k) = 1$, define $F(x) = \sum_{x_k \leq x} p(x_k)$. Then (i) $0 \leq F \leq 1$ and $F(x) = 0$ for $x < \min x_k$; (ii) F is non-decreasing (adding non-negative terms); (iii) $F(x_k) - F(x_k^-) = p(x_k)$, a jump; (iv) because of “ $\leq$ ”, $F(x_k) = F(x_k^+)$, so F is right-continuous, but $F(x_k^-) \neq F(x_k)$ when $p(x_k) > 0$, so it is not left-continuous. Q.54

I. Normal and Cauchy (t_1) laws

Densities. $g(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$. The Student- t_ν density is $\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} (1 + \frac{y^2}{\nu})^{-\frac{\nu+1}{2}}$; for $\nu = 1$, with $\Gamma(1) = 1$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$:

$$h(y) = \frac{1}{\pi(1+y^2)}, \quad H(y) = \int_{-\infty}^y h = \frac{1}{2} + \frac{\arctan y}{\pi}. \quad (0.36)$$

Symmetry. For an even pdf, $F(-y) = 1 - F(y)$ and $F(0) = \frac{1}{2}$. Hence $G(c) < H(c) \iff G(-c) > H(-c)$: exactly one of the statements $G(c) < H(c)$ and $G(-c) < H(-c)$ can hold.

Characteristic functions (Fourier transforms of the pdf). For the normal law, completing the square,

$$\varphi_Z(\omega) = \int e^{i\omega z} g(z) dz = e^{-\omega^2/2}, \quad (0.37)$$

and for the Cauchy law, by residues (closing the contour in the upper half-plane for $\omega > 0$, pole at $y = i$),

$$\varphi_Y(\omega) = \int \frac{e^{i\omega y}}{\pi(1+y^2)} dy = e^{-|\omega|}. \quad (0.38)$$

The Cauchy law has $|\omega|$ in the exponent (non-differentiable at 0), which corresponds to the $1/y^2$ tail and non-existent mean, whereas the normal law has a Gaussian tail.

Uniqueness of the crossing point $c > 0$. Setting $g(c) = h(c)$:

$$\frac{e^{-c^2/2}}{\sqrt{2\pi}} = \frac{1}{\pi(1+c^2)} \iff q(c) := \sqrt{\frac{\pi}{2}} (1+c^2) e^{-c^2/2} = 1. \quad (0.39)$$

Now $\frac{d}{dc} [(1+c^2)e^{-c^2/2}] = c(1-c^2)e^{-c^2/2}$, so q increases on $(0, 1)$ from $\sqrt{\pi/2} \approx 1.25 > 1$ to $q(1) = 2\sqrt{\pi/2} e^{-1/2} \approx 1.52$, then decreases to 0. Hence $q = 1$ has exactly one root $c > 0$, with $c > 1$; numerically $q(1.85) \approx 1.002$, $q(1.86) \approx 0.992$, so $c \approx 1.85$. Then $H(-c) = \frac{1}{2} - \frac{\arctan 1.85}{\pi} \approx 0.158$ and $G(-c) \approx 0.032$. Q.28

J. Poisson law, Central Limit Theorem

$N \sim \text{Poisson}(\mu)$: $P(N = k) = e^{-\mu} \mu^k / k!$, $N = N = \mu$, and the characteristic function is

$$\varphi_N(\omega) = \sum_k e^{i\omega k} \frac{e^{-\mu} \mu^k}{k!} = \exp(\mu(e^{i\omega} - 1)). \quad (0.40)$$

Hence $\varphi_{N_n} = \varphi_{N_1}^n$ for $\mu = n$: N_n is a sum of n i.i.d. Poisson(1) variables. The Central Limit Theorem (Fourier-transform proof: $\varphi_{(S_n - n\mu)/\sqrt{n}\sigma}(\omega) \rightarrow e^{-\omega^2/2}$) gives

$$\frac{N_n - n}{\sqrt{n}} \xrightarrow{d} N(0, 1) \Rightarrow P(N_n \leq n) = P\left(\frac{N_n - n}{\sqrt{n}} \leq 0\right) \rightarrow \Phi(0) = \frac{1}{2}. \quad (0.41)$$

Q.45

6. OPTIMISATION AND COMPUTER-SCIENCE FACTS

A. Stochastic gradient descent

For loss $f_w(x)$ and learning rate η :

$$w_{t+1} = w_t - \eta \left. \frac{\partial f_w(x)}{\partial w} \right|_{w_t}, \quad (0.42)$$

and for $f_w(x) = wx$, $\frac{\partial f}{\partial w} = x$, so $w_{t+1} = w_t - \eta x$. Q.29

B. Bubble sort pass and inversions

An inversion is a pair $i < j$ with $a_i > a_j$. Swapping an adjacent out-of-order pair (a_k, a_{k+1}) removes exactly that inversion and changes the relative order of no other pair, so the inversion count drops by exactly 1 per swap. A fully sorted array has 0 inversions. Therefore the total number of swaps performed until the array is sorted equals the initial inversion count. For $[5, 3, 4, 1, 2]$: $4 + 2 + 2 + 0 + 0 = 8$ inversions (counted by the smaller-indexed element). Q.58

C. Python semantics used

- Default parameter values are evaluated once, at function definition time; a mutable default is shared by all calls that omit the argument.
- Each call of an enclosing function creates a new local scope; an inner function (closure) keeps a reference to its own enclosing variables, so two closures from two calls share nothing.

Q.16, Q.50